



Correlation of autophagy and Alzheimer's disease with special emphasis on the role of phosphodiesterase-4

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Abstract

Autophagy disruption is important in Alzheimer's disease (AD) as it prevents misfolded proteins from being removed, which leads to the accumulation of amyloid plaques and neurofibrillary tangles (NFTs). Restoring autophagy improves neuronal survival and cognitive function, according to experimental models. In AD models, mTOR inhibition and AMPK activation enhance synaptic plasticity and lessen learning deficits. Inhibitors of phosphodiesterase-4 (PDE4) improve cognition and reduce neuroinflammation via altering cyclic adenosine monophosphate (cAMP) transmission. Furthermore, autophagic-lysosomal clearance is encouraged by upregulating transcription factor EB (TFEB), which lessens the pathogenic damage linked to AD. These results point to autophagy modification as a promising therapeutic approach, with the mTOR, AMPK, cAMP, and TFEB pathways being possible targets for drugs. Though much evidence is based on animal studies, these findings provide valuable insights into autophagy's role in AD pathology, offering promising directions for future research and drug development.

Keywords Alzheimer's disease (AD) · Autophagy · Cyclic adenosine monophosphate (cAMP) · mTOR · Phosphodiesterase-4 (PDE4)

Introduction

Alzheimer's disease (AD) is a gradual and irreversible neurological condition affecting large areas of the cerebral cortex and hippocampus (Masters et al., 2015). It is the most prevalent kind of dementia, marked primarily by rapid loss of memory and cognitive impairment, which later affects behavior, communication, visual orientation, and motor function (DeTure and Dickson 2019). Over time, AD diagnosis has evolved from being solely based on pathology to a clinical, exclusionary approach (Scheltens et al. 2021). The accumulation of amyloid plaques and the presence of neurofibrillary tangles in a patient's brain distinguishes the disease neuropathologically. Amyloid plaques are classified morphologically into two types: diffuse and neuritic. Even in those who do not have dementia, the existence of both kinds may suggest an increased risk of developing dementia.

While neuritic plaques are associated with cognitive loss, the relationship between diffuse plaques and cognitive decline is less evident. Plaque development is initiated by the deposition of insoluble A β amyloid (Kondaveeti et al. 2024). Dystrophic axons and dendrites are observed in neurotic plaques, with the latter containing pathogenic bundles of tau proteins identical to those found in neuronal perikaryon creating neurofibrillary tangles (NFTs). Tau is a cytoskeletal protein that is essential for the stability of microtubules in the neuronal cytoskeleton. Tau undergoes aberrant phosphorylation in the presence of neuritic plaques, likely affecting microtubule interaction and promoting tau aggregation into paired helical filament (Tausif et al. 2025). This aggregation is expected to impair the neuron's capacity to sustain large dendritic and axonal arborizations, resulting in synaptic connection loss and neuronal death (Caselli et al. 2017). AD causes a gradual loss of synapses in the cerebral cortex and hippocampus, which impairs memory and worsens cognitive abilities (Scheff et al. 2006). The majority of AD patients have the sporadic variety, which has a late start (usually between the ages of 80 and 90). This variation is caused by a failure to remove the amyloid- β (A β) peptide from the interstitial spaces of the brain (Divya et al. 2025). Alzheimer's disease

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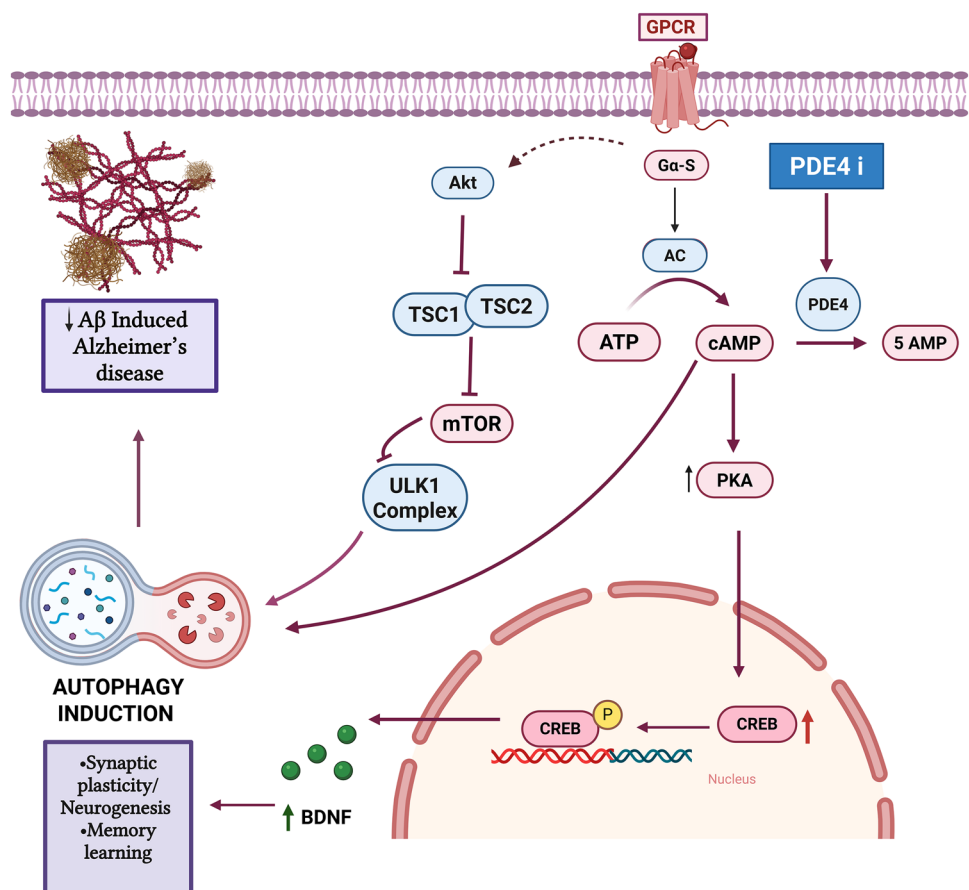
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involves multiple factors, including cognitive decline, hormonal influences, nutrient imbalances, and potential treatments such as traditional medicinal plants (Singh et al. 2024; Tripathi et al. 2024; Yadav et al. 2024). Numerous genetic risk factors for sporadic AD have been found. In a tiny percentage of instances (1%), individuals who inherit mutations impacting a processing gene have onset at a younger age, average about 45 years. Using cerebrospinal fluid biomarkers and PET scans, amyloid- β buildup may now be detected in preclinical and prodromal phases. While there are some authorized drugs that can help with specific AD symptoms, none of them can change the underlying disease processes (Masters et al., 2015). With approximately 50 million affected individuals worldwide, AD imposes a significant burden on individuals, families, and the global economy, with estimated annual costs of US\$1 trillion (Breijyeh and Karaman 2020). In India alone, dementia has affected more than 4 million people (Shanmugam and Gopal 2020), and by 2030, the load is predicted to reach 7.5 million (Nangare and Patil 2022). Alzheimer's disease affects brain function, and researchers are studying ways to improve memory and protect brain cells using cell signaling pathways (Ramakrishna et al. 2023; Srivastava et al. 2019; Tripathi et al. 2019). Researchers are exploring treatments that target harmful

protein buildup and energy imbalances in brain cells (Rai et al. 2020; Ramakrishna et al. 2024).

Research suggests a significant correlation between autophagy dysfunction and AD, enhanced autophagy promotes the clearance of amyloid-beta plaques and tau aggregates, reducing neurotoxicity and alleviating symptoms associated with AD. Autophagy is a set of actions that culminate in the creation of a mature autophagosome. These steps include (a) the development of a phagophore, a double-membrane structure, (b) the elongation of this lipid-based membrane, and (c) the sequestration of intracellular cargo. Subsequently, macromolecules are degraded and recycled by autolysosomes, which result from the fusion of autophagosomes and lysosomes. This crucial process helps maintain the cell's energy homeostasis. Autophagy has been linked to a number of biological activities, including mammalian development, preventing diseases, and lengthening of life (Komatsu et al. 2006). Autophagy has been extensively studied, although little is still known about how it functions in neurons. Recent research has demonstrated that autophagy is required for the clearance of amyloid- β proteins from the brain, which helps to protect neurons as described in Fig. 1 (Hara et al. 2006; Komatsu et al. 2006; Rubinsztein et al. 2007).

Fig. 1 Regulation of Autophagy via the cAMP Pathway. This highlights important chemical interactions and shows how the cAMP pathway regulates autophagy. PKA is activated by elevated cAMP levels and phosphorylates key autophagy-related proteins, hence regulating autophagy. This regulation of autophagy is important in Alzheimer's disease, as increasing autophagy by modulating cAMP may aid in the removal of misfolded proteins and lessen neurodegeneration



Autophagy can be regulated through caloric restriction, exercise, pharmacological agents (e.g., rapamycin), and modulating signaling pathways like mTOR and AMPK. PDE4 inhibitors enhance autophagy by increasing cAMP levels, which activate pathways promoting autophagic activity and reducing neurodegeneration. There are 11 isoenzyme families that comprise phosphodiesterases (PDEs) (PDE1 to PDE11), and each of these groups has distinct drug sensitivity patterns, tissue and subcellular space distribution, substrate specificities, and regulating mechanisms (Abusnina and Lugnier 2017). PDEs are in charge of breaking down the 3'-phosphodiester link in cyclic AMP and cyclic GMP, which are essential signal-transduction molecules that control cellular activity like gene expression, inflammation, and synaptic transmission. (Boswell-Smith et al., 2006). PDE4 inhibition has been related to a transition in the hippocampus between memory extinction and strengthening, and combining retrieval with dorsal hippocampus PDE4 inhibition can pharmacologically shift the behavioral result of the recall of memories from extinction to strengthening (Roesler et al. 2014). Activation of cAMP signaling enhances memory while decreasing inflammatory and apoptosis responses. It is unknown, however, whether blocking PDE4, an essential control of internal cAMP levels, impacts the neuroinflammatory and apoptosis linked to dementia and AD, and if these reactions are associated with memory problems induced by cAMP signaling (C. Wang et al. 2012). Evidence currently available suggests that the transcription factor cAMP-response element-binding protein (CREB), which in turn activates gene expression and essential protein synthesis for long-term memory formation, is mediated by cAMP-dependent activation of protein kinase A (PKA) or mitogen-activated protein (MAP) kinase (Rose et al. 2005). Moreover, the previous studies have indicated that the retrograde transport, i.e., movement of molecules, vesicles, or organelles from the cell's periphery back toward the nucleus or the cell body of BDNF-activated tropomyosin-related kinase B (TrkB) receptors plays a role in the formation of autophagosomes, enhancing neuronal complexity and helping to prevent neurodegeneration in vivo. Additionally, BDNF expression is regulated by sortilin-mediated trafficking and lysosomal degradation as described in Fig. 1 (L. Zhang et al. 2021).

This review's goal is to provide an overview of how autophagy regulation and AD are related and it also emphasizes PDE4's function in the regulation of autophagy pertaining to AD and its inhibition being a therapeutic target for AD.

Pathophysiology of AD

AD is a prevalent age-related neurodegenerative condition and the most common form of dementia worldwide. It is

characterized by the presence of extracellular amyloid- β (A β) plaques and intracellular neurofibrillary tangles (NFTs) composed of hyper-phosphorylated tau protein (Masters et al., 2015). The accumulation of the toxic A β 42 and A β 40 forms, coupled with impaired clearance mechanisms, drives the formation of A β plaques in AD pathogenesis (McGowan et al. 2005). Although the majority of AD cases (approximately 90%) are sporadic and not linked to inherited genetic mutations related to amyloid processing genes, they still exhibit β -amyloid plaques, sharing similarities with familial AD. Several common risk factors associated with cardiovascular disease and dementia, such as aging, hypertension, hypercholesterolemia, and diabetes, contribute to AD risk. Genetic polymorphisms in certain risk factor genes significantly increase the likelihood of developing AD, with the APOE ϵ 4 allele being the most prevalent, contributing to about 50–60% of AD cases (Saunders et al. 1993). Neurofibrillary tangles (NFTs) represent another critical pathological feature in AD. These intracellular protein aggregates exert cellular toxicity, disrupting signaling within cells and leading to neuronal death. Interestingly, research suggests that tau protein, rather than the tangles themselves, maybe the mobile element responsible for the spread of NFT pathology throughout the brain, even in regions that do not display NFTs at the time of death (DeVos et al. 2018; Holmes et al. 2014). Genetic polymorphisms related to the autophagy-lysosomal pathway have been identified as disease risk factors for late-onset AD, indicating a link between autophagy and AD (Scheltens et al. 2021). One of the primary methods through which the cell destroys aberrant proteins is autophagy. Thus, the removal of such protein clumps may be accomplished by employing autophagy pathways (Miners et al. 2017). Figure 2 illustrates the contrasting cellular characteristics between AD affected brains and those treated with autophagy-inducing therapies.

Link between autophagy regulation and AD

Autophagy is a process that involves the formation of autophagosomes, sequestering vesicles crucial for this process, and most of the proteins involved in autophagy are primarily associated with vesicle formation (Yorimitsu and Klionsky 2005). Although mammalian cells were the first to exhibit autophagy in the 1960s, the underlying molecular mechanism has only recently begun to be elucidated (Thumm et al. 1994; Tsukada and Ohsumi 1993). Our knowledge of the molecular foundation of autophagy has been greatly improved by the identification of autophagy-related (ATG) genes, notably in yeast (Klionsky et al. 2003). Deficits in A β clearance, tau-related pathology, neuroinflammation, mitochondrial dysfunction, and compromised proteostasis are among the pathways linked to the development of AD. Unlike conventional neuroprotective approaches

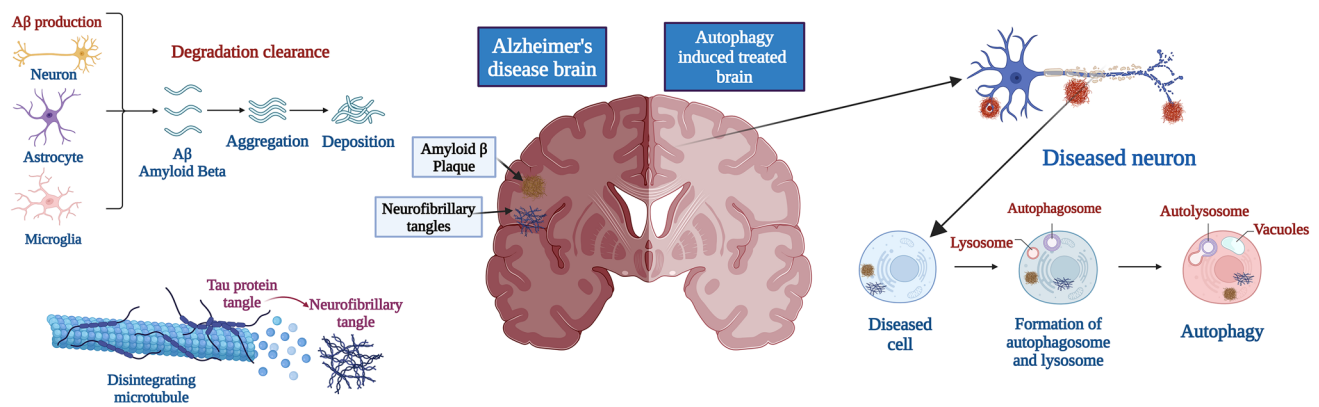


Fig. 2 Comparative analysis of AD and autophagy-enhanced brain-pointing up significant variations in cellular functions. Toxic protein aggregates build up as a result of defective autophagy in AD, which exacerbates neurodegeneration. On the other hand, improved

autophagy supports neuronal health by encouraging the removal of damaged organelles and misfolded proteins. Comprehending these distinctions could facilitate the creation of AD treatments that target autophagy

that mainly target neurotransmitter systems or individual pathogenic proteins, autophagy modulation provides a dual clearance pathway for both tau and amyloid-beta disease (Barmaki et al. 2023). The necessity of precisely regulating autophagic activity to preserve neuronal health is highlighted by the fact that excessive autophagy activation in neurons can be harmful and can cause cell death. Targeting important regulatory molecules and pathways, including mTOR inhibitors or AMPK activators, is one strategy to manage autophagy and attain a balanced autophagic response (Button et al. 2015).

Reduced autophagy has been linked to the development of AD along with various progressive neurodegenerative disorders (Metaxakis et al. 2018), which frequently involves altered substrate orientation and autophagosome development as well as decreased expression of key autophagy receptors. Notably, a number of autophagy-related topics, including maturation, degradation, and lysosomal alterations, have drawn a lot of attention (Son et al. 2012). The ULK1 complex, which is made up of the scaffold proteins Atg13, Atg101, ULK1, and FIP200, is an essential starter of mammalian autophagy. The mTORC1 pathway, one of the stress signaling pathways, among others, has an impact on the ULK1 complex's activity. The energy level-dependent sensor AMPK (5' adenosine monophosphate-activated protein kinase) is activated when intracellular amino acids are reduced as a result of fasting. mTORC1 is separated from ULK1 by AMPK's phosphorylation of mTORC1. As a result, the transcription factor EB (TFEB) therefore enters the nucleus and encourages the overexpression of genes related to lysosomes and autophagy. Atg13, Atg101, and FIP200 combine to generate a combination with ULK1 (Ganley et al. 2009). The AMPK/mTOR signaling pathway is thought to

be activated by mitochondrial malfunction brought on by aging, which results in aberrant energy metabolism and the etiology of AD (Ju et al. 2011). A promising strategy for treating AD involves enhancing the clearance of Tau protein by targeting TFEB-mediated autophagy (H. Wang et al. 2016).

Studies suggest that inducing autophagy in the context of impaired autophagic vesicle clearance may worsen AD pathology, which has implications for using autophagy modulation as a potential therapeutic approach in neurodegenerative diseases (Boland et al. 2008). Recent evidence indicates that enhancing autophagy holds promise as a therapeutic strategy to counteract the toxicity caused by misfolded proteins in age-related neurodegenerative conditions (Rubinshtein et al. 2007). Autophagy is regulated by signaling pathways found in neurons, and it is essential for the development of oligodendrocytes, myelination, and synaptic plasticity (Kesidou et al. 2013; J.-A. Lee 2012). The functional state of autophagy is crucial for the overall health of organisms, but aging is associated with a noticeable decline in autophagic activity (Del Roso et al. 2003). The role of the cAMP signaling system in long-term memory formation is widely recognized. When cAMP-dependent protein kinase A (PKA) is activated by allosteric interaction with cAMP, it phosphorylates CREB at serine 133. Following that, phosphorylated CREB works with the transcription coactivator CREB-binding protein to promote transcription and translation of certain CREB target genes. These genes are essential for synaptic plasticity, which is required for the development of long-term memory (Y. Chen et al. 2012; Del Roso et al. 2003). In Table 1, we present a detailed overview of PDE4 drugs with its dosing guidelines in both human and animal contexts.

Table 1 Comprehensive dosing overview of PDE 4 drugs: human and animal comparisons

Compounds	Human Dose for COPD	Animal dose for COPD	Animal dose for AD	Reference
Roflumilast	250 or 500 mcg tablet once daily	1 or 5 mg/kg	5 or 10 mg/kg	(Hatzelmann et al. 2010; H. Wang et al. 2020a, b)
Apremilast	30 mg twice daily	10 mg/kg or 20 mg/kg	5 mg/kg	(Al-Harbi et al. 2023; Gomaa et al. 2023)
Rolipram	–	0.25 mg/kg	0.03 mg/kg	(Saletu et al. 1992; C. Wang et al. 2012)
Cilomilast	15 mg twice daily	3–30 mg/kg	0.1–10 mg/kg	(Kobayashi et al. 2011)

Amyloid β precursor protein

The formation of autophagosomes in dystrophic neurites within AD brains is an indication of the dysregulation of autophagy in AD. However, it is yet to be determined whether this dysregulation is caused by increased autophagy initiation or decreased lysosomal clearance (M. J. Lee et al. 2013). Autophagy has been associated with two key biochemical hallmarks of AD: Amyloid-beta ($A\beta$) and Tau. Autophagosomes in AD contain $A\beta$, and regulating autophagy genetically or pharmacologically affects $A\beta$ pathology in AD model mice. Recent studies have clarified the function of autophagy in $A\beta$ metabolism, demonstrating how it affects the production and secretion of $A\beta$ plaques (Nilsson et al. 2013). Reduced levels of the autophagy-related protein BECN1 have been linked to increased $A\beta$ accumulation, while enhanced BECN1 levels reduce $A\beta$ accumulation (Pickford et al. 2008). Autophagy and the endosomal/lysosomal system are upregulated early, highlighting their critical role in mitigating cellular stress and potentially delaying the progression of AD (Ditaranto et al. 2001). In early AD condition, autophagy is upregulated as a compensatory mechanism to enhance the clearance of amyloid-beta and tau aggregates, aiming to mitigate neurodegeneration.

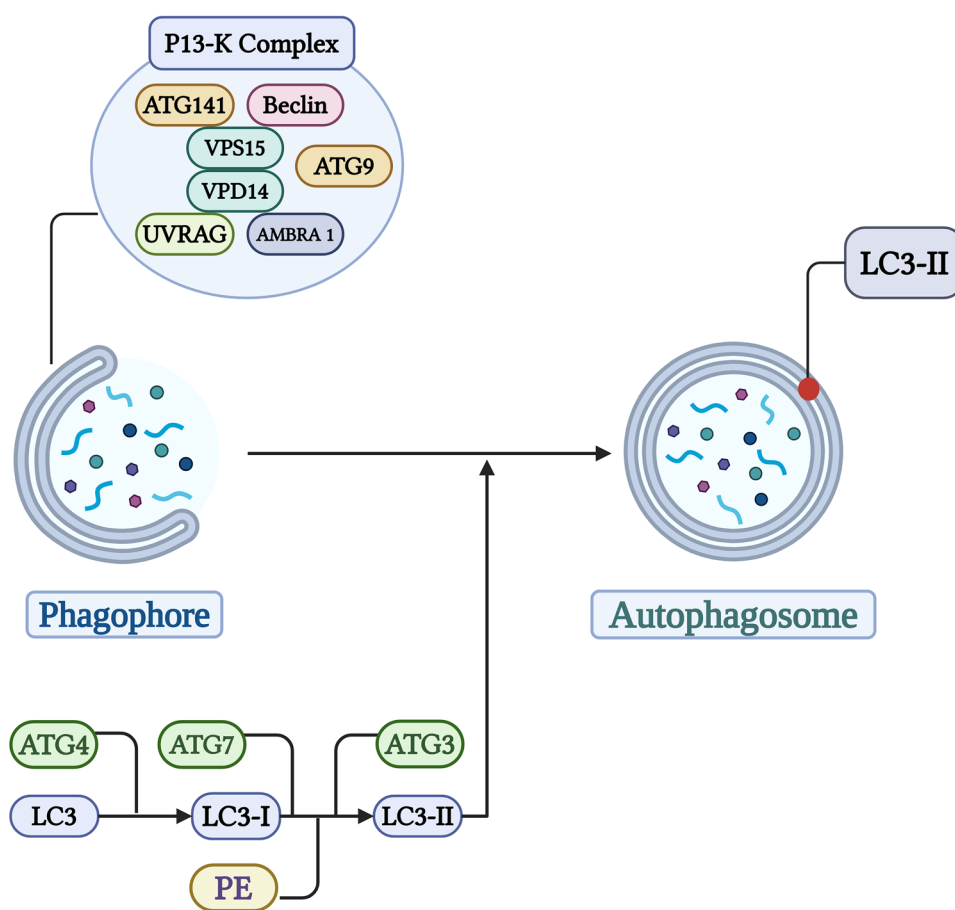
In AD, one extensively studied pathology involves the increased levels and aggregation of $A\beta$, leading to plaque formation. Mutations to the APP, PS1, or PS2 genes cause $A\beta$ overproduction in autosomal-dominated family, especially the aggregation-prone $A\beta$ 42 and $A\beta$ 43 (Bettens et al. 2013; Lambert et al. 2013). In addition, additional risk genes for AD have been discovered, with various roles in the immune system, endocytosis, and synaptic activity. Mouse models mimicking impaired lysosomal clearance accumulate autophagosomes and show increased APP-CTF and $A\beta$ levels (Boland et al. 2010). Autophagy is induced not only by starvation but also by oxidative stress or proteasome inhibition, which can enhance cellular resilience by promoting the clearance of amyloid-beta ($A\beta$) and reducing lysosomal accumulation (Agholme et al. 2012; Zheng et al. 2011). An excessive amount of $A\beta$ may have deposited in mice neurons when autophagy-related genes, such as Atg5 and Rubicon, are deleted, whereas enhancing LC3-associated

endocytosis improves $A\beta$ clearance and lessens AD symptoms as described in Fig. 3 (Heckmann et al. 2019). Hippocampal neurons with mutant APP and $A\beta$ may exhibit impaired mitophagy and mitochondrial fragmentation, which may impair neuronal function (Sugarman et al. 2019). There is proof that soluble APP (sAPP) has a neuroprotective effect and performs similarly to growth factors (Del Prete et al. 2016). Additionally, mitochondria-associated membranes (MAMs) are sites where the amyloid precursor protein (APP) and its proteolytic components are processed subcellularly, which may aid in the etiology of AD (Lambert et al. 2013). Axon degeneration and pruning, cell adhesion, apoptosis, synaptic dysfunction in AD, neuronal survival, differentiation, and transport vesicles to synaptic sites are all dependent on APP proteins (Lopez Salon et al. 2003).

Tau protein

Tau protein formation, driven by phosphorylation and subsequent aggregation into neurofibrillary tangles, is influenced by autophagy, which plays a crucial role in the clearance of these toxic aggregates in AD. The ubiquitin–proteasome system (Ups) is probably the main method through which endogenous tau is broken down (Brown et al. 2005). Rapamycin and other autophagy promoters have been discovered to accelerate the breakdown of insoluble tau and guard against tau toxicity in *Drosophila* (Berger et al. 2006). According to research, the acetyltransferase CREB-binding protein (CBP) acetylates a tau fragment that contains a microtubule-binding domain, which causes the fragment to aggregate more frequently (Cohen et al. 2011). Acetylated tau has been suggested to inhibit chaperone-mediated autophagy (CMA), favoring macroautophagy and microautophagy of endosomes as a means of tau breakdown. Contradictory results, however, claim that acetylating its entire length of tau via acetyltransferase p300 lowers the accumulation of tau and that the histone deacetylase (HDAC6) can reverse this effect (Cook et al. 2014). In the brains of AD patients, phosphorylated tau has been seen to co-localize with an accumulation of autophagic vesicles, suggesting a connection between tauopathy and defective autophagy (Piras et al. 2016). Tau can be phosphorylated when mTOR activity is elevated, but tau can also be degraded when mTOR

Fig. 3 Regulation of autophagy initiation by PI3-K complex and LC3-II. The figure highlights important molecular connections by showing how the PI3-K complex and LC3-II regulate autophagy initiation. By triggering downstream signaling pathways that control the development of autophagosomes, the PI3-K complex plays a critical role in the early phases of autophagy. Cargo sequestration and autophagosome membrane growth depend on LC3-II, a lipidated version of LC3. Neurodegenerative illnesses like Alzheimer's are associated with dysregulation of this process, where toxic protein buildup is a result of compromised autophagy



activity is inhibited, as it has been demonstrated in a number of AD animal models (Caccamo et al. 2013). Autophagy inducers increase total and phosphorylated tau secretion in a human neuroblastoma cell line overexpressing human wild type full-length tau (hTau), and this effect is inhibited by beclin1 knockdown or autophagy inhibitors (Kang et al. 2019).

Role of mTOR in autophagy regulation

The protein kinase mTOR, which is highly expressed in the brain, is essential for controlling the ratio of protein synthesis to breakdown. In response to growth hormones and guidance cues, it functions as a crucial signal for neuronal survival and division throughout embryonic development. Although their precise processes are still not completely understood, coordinated mTORC1 and mTORC2 activities are crucial for appropriate neuronal and brain development (Meijer et al. 2015). As a serine–threonine kinase, mTOR regulates different elements of mammalian cell activity in response to intracellular and extracellular inputs, evaluating cell resources and health to react to external stimuli (Loewith et al. 2002; Perluigi et al. 2015; Switon et al. 2017). Due to its role in regulating translation and autophagy, it

has been extensively investigated, and scientists have looked into the curative effects of rapamycin, an mTOR inhibitor (Bové et al. 2011; Lipton and Sahin 2014; Narayanan et al. 2024). Rapamycin therapy or a partial genetic mTOR concentration lowering has been shown to enhance memory and learning issues in models for AD. The study reveals that naturally secreted amyloid- β enhances mTOR activity through a PRAS40-mediated mechanism, suggesting a novel link between amyloid signaling and mTOR regulation. Inhibition of mTOR by rapamycin, on the other hand, effectively reverses cognitive deficits and reduces amyloid- β levels in a mouse model of AD, indicating potential therapeutic avenues (Caccamo et al. 2011; Spilman et al. 2010).

PI3-K/Akt pathway

Pathological forms of tau and phospho-tau proteins are correlated with the accumulation of autophosphorylated mTOR at Ser2481 site and phosphorylated at the Ser2448 site via phosphoinositide-3 kinase (PI3K) signaling pathway (X. Li et al. 2005). As it contributes to the autophagy failure associated with AD pathogenesis, mTOR signaling inhibition has positive impacts on both the pathophysiology and behavioral outcomes in a number of AD mice models (Caccamo

et al. 2011; Nixon 2013; S. Zhang et al. 2010). Rapamycin has generated interest due to its potential as an "anti-aging" medication due to its capacity to prolong lifespan and delay age-related impairments (Holtzman et al. 2011).

The importance of the mTOR signaling pathway in the progression of AD has been backed up by studies that indicate that decreased clearance of abnormal conformations significantly contributes to synaptic deviations (Guo et al. 2011). Cognitive severity and mTOR hyperactivity are linked in AD patients (Tramutola et al. 2015). In the AD brain, PI3-K/Akt axis-mediated mTOR upstream signaling is also reduced as described in Fig. 3 (Martín et al. 2001; Wei 2002). Cognitive impairment and the progression of AD can also be caused by excessive mTOR activity (Meijer and Codogno 2006). Downstream mTOR components, such as activated p70S6K, co-localize to hyper-phosphorylated tau depositing in the brains of AD patients, demonstrating their involvement in tau accumulation (An et al. 2003). SIRT1 can prevent neuronal degeneration induced by an exposure by inhibiting mTOR as well as p70S6 kinase (p70S6K) (Guo et al. 2011).

Temsirolimus, an mTOR inhibitor approved by the FDA for renal cell carcinoma treatment, has shown promising effects in AD models, improving spatial learning and promoting A β clearance (Jiang et al. 2014). In addition, it can enhance motor impairments in mutant tau animal models and increase macroautophagy and inhibiting mTOR, Akt, and p70S6K signaling (Frederick et al. 2015).

AMPK activation

Three subunits form the heterotrimeric complex known as AMPK (AMPK α , AMPK β , and AMPK γ), which exists in seven isoforms in eukaryotic cells, capable of forming twelve potential AMPK complexes (Calabrese et al. 2014; Kurumbail and Calabrese 2016). The expression of AMPK plays a critical role in neurotoxicity-related diseases, as it is highly expressed in the brain, protecting neurons during low-energy periods and preventing or reducing neuronal death (Ju et al. 2011). AMPK promotes cellular autophagy and inhibits the anabolism of sugars and lipids, facilitating catabolism, while mTOR, on the other hand, functions oppositely. The balance between mTOR and AMPK activities greatly influences cellular energy metabolism homeostasis. The AMPK/mTOR metabolic pathway is highly active in the brains of AD patients (Majd and Power 2018). By blocking the activation of the AMPK/mTOR signaling pathway, rapamycin is known to lower the incidence of AD and improve lesions associated with AD in type 2 diabetic rat hippocampus (Sun et al. 2019). AMPK is triggered in response to variations in intracellular AMP/ATP ratios when cells need energy, and it immediately phosphorylates TSC2 to prime it for later GSK3-mediated phosphorylation, which inhibits

mTOR signaling (Inoki et al. 2006). The comparative activity of AMPK and mTOR determines the degree of activation of cell autophagy in different energy conditions. In the presence of sufficient nutrition, ULK1 is phosphorylated by mTORC1, which prevents AMPK from activating ULK1 and inhibiting autophagy. However, AMPK becomes active, mTOR is inhibited, and autophagy is initiated in cells that lack energy (Y. Wang et al. 2020a, b).

The combination of Unc-51-like kinase 1 (ULK1) and vacuolar protein sorting (Vps34) is a critical regulator of autophagy initiation and progression. Autophagy can be initiated via ULK1 and AMPK activation, increased PI3P levels, or altered intracellular calcium signals, in addition to mTOR-dependent mechanisms (Johnson et al. 2012). AMPK inhibits mTORC1 by lowering Ser 757 phosphorylation, whereas mTOR inhibits ULK1 activity by phosphorylating it. This results in less damage from mTOR and activation of the AMPK/ULK1 signaling pathway, which promotes normal autophagy initiation and slows down the aging of organisms (Ge et al. 2022). Resveratrol has been shown to activate key physiological sensors/effectors such as sirtuin 1 (SIRT1), AMP-activated protein kinase (AMPK), and peroxisome proliferator-activated receptor co-activator-1 (PGC-1 α) (Um et al. 2010).

Upregulation of TFEB

Delivery of TFEB, an affiliate of the MiT family of bHLH-Zip transcription elements, has proved effective in lowering neurofibrillary tangle and alleviating behavioral abnormalities, cognitive impairments, and neurodegeneration in the rTg4510 animal model of tauopathy (Steingrímsson et al. 2004). By encouraging the formation of lysosomes, autophagosomes, and fusion with lysosomes, TFEB enhances autophagic flow and facilitates the effective degradation of complex molecules. When activated, TFEB directly attaches to promoter sequences to increase the expression of genes linked to autophagy and lysosomes. Dysregulation of TFEB activity may be a factor in the onset of a number of illnesses, such as inflammatory diseases, cancer, neurodegenerative diseases, and hepatic steatosis (Franco-Juárez et al. 2022).

When TFEB is overexpressed, hazardous p-Tau species can be selectively targeted without harming healthy neurons (Polito et al. 2014). This selectivity for the TFEB-mediated breakdown of harmful Tau molecules has also been demonstrated in a genetic neuron-specific TFEB overexpression paradigm (H. Wang et al. 2016). However, reports of effective small-molecule TFEB activators are rare, and their use is constrained by the blood–brain barrier's limited permeability (Yang et al. 2022). The main factor regulating TFEB activity is phosphorylation, which is controlled by a number of phosphatases and kinases, including mTORC1, which is

the main kinase controlling TFEB nuclear translocation. At a number of serine residues, including S211, S142, and S122, mTORC1 phosphorylates TFEB (Puertollano et al. 2018).

Protein kinase C (PKC) stimulate TFEB translocation into the nucleus by blocking GSK3-induced phosphorylation of TFEB (Y. Li et al. 2016; Napolitano et al. 2018). GSK3 phosphorylates TFEB at Ser134 and Ser138. In an mTOR-independent way, Akt phosphorylates TFEB at Ser467, encouraging its cytoplasmic increase (Palmieri et al. 2017). A recent study has shown that inhibiting GSK3 selectively activates TFEB, enhancing lysosomal biogenesis and lowering levels of APP and APP-carboxy-terminal segments, lending credence to the concept that TFEB-induced autophagic–lysosomal clearance ameliorates pathophysiological deficits associated with AD (Parr et al. 2012). PPAR/TFEB inhibition in astrocytes has also demonstrated positive effects in AD. The activation of PPAR by aspirin increases TFEB levels and lysosomal biogenesis in 5XFAD mouse astrocytes. The discovery that removal of PPAR in 5XFAD mouse contributes to plaque pathogenesis and cannot be restored by aspirin treatment emphasizes the relevance of aspirin-regulated PPAR-TFEB in astrocytes for amyloid plaque clearance (Chandra et al. 2018). Acetylation, in addition to phosphorylation, controls the activity of TFEB. Suberoylanilide hydroxamic acid (SAHA), which inhibits histone deacetylase, causes TFEB to be acetylated at Lys91, Lys103, and Lys430, increasing its transcriptional activity without reducing nuclear accumulation (J. Zhang et al. 2018). The distribution of TFEB within cells is significantly influenced by phosphorylation. The nuclear translocation of TFEB is inhibited by GSK-3 β phosphorylation at serine 134 and 138, which causes it to localize on the surface of lysosomes where it interacts with mTORC1. Reduced lysosomal localization and increased nuclear localization are the effects of TFEB S134A and S138A double mutations (Nabar and Kehrl 2017). In terms of AD, it has been shown that overexpressing TFEB can reduce APP/A β or microtubule-associated protein tau (MAPT) pathogenesis. Exogenous activity of the TFEB expression, produced by AAV administration in the hippocampus astrocytes and nerve cells of APP/PS1 mice, increases the degradation of APP and APP C-terminal fragment, resulting in a reduction in A β -generation (Xiao et al. 2014). The detailed autophagy regulations involved in AD are illustrated in Fig. 4.

Role of mTOR-independent pathway in autophagy regulation through upregulation of cAMP pathway

cAMP is a secondary universal messenger that is change in response to the binding of G-protein-coupled receptor (GPCR) that is connected to an adenylyl cyclase (AC) enzyme. This enzyme promotes the cyclization of adenosine triphosphate (ATP), which results in the production

of cAMP (Halls and Cooper 2017). cAMP levels are often altered in AD, with variations toward decrease, elevation, or stability depending on the particular expression and activity of AC and PDE in different brain areas. This diversity may contribute to the complex pathophysiology of AD. Notably, basal, Gs-mediated (a type of G-protein that stimulates adenylyl cyclase (AC) activity), and forskolin-induced AC activity are reduced in the hippocampus and temporal cortex, owing to decreased expression of AC1 and AC2, among other AC isoforms. Interestingly, Gs-mediated AC activity decreases specifically in the cerebellum and occipital cortex, showing regional subtleties in the control of cAMP signaling pathways (Di Benedetto et al. 2021; Kelly 2018). The creation of multivalent complexes with A-kinase anchoring protein (AKAP) is critical for effective spatial and temporal modulation of PKA-mediated signaling, which is critical for maintaining endothelial barrier function. The activation of Rac1, a crucial signaling molecule involved in barrier stability by PKA, takes place in close proximity to the anterior junctions and the cortical actin cytoskeleton. This specificity is related to Rac1's attachment to AKAP220, which ensures compartmentalized control of this critical signaling pathway (Radeva et al. 2014). PKA is made up of two catalytic and two regulatory components that work together to generate a dynamic signaling complex. This cyclic nucleotide subsequently interacts with PKA's regulatory elements (PKA-R), causing PKA-R to dissociate from the catalytic components. This coordinated dissociation liberates the PKA catalytic components, allowing them to perform the critical duty of phosphorylating CREB/PKA dysfunction is being investigated in animal models of AD, with A β -induced oxidative stress being identified as one initiating mechanism (Du et al. 2014). Furthermore, A β -independent mechanisms, such as direct PKA pathway blockage by γ -secretase (BACE)1 (Y. Chen et al. 2012), contribute to the thorough analysis of PKA-related modifications in the complicated pathophysiology of AD. The activation of Epac by cAMP allows for the repair of barrier characteristics in the blood–brain barrier (BBB) following a breakdown. Despite the well-established role of cAMP in regulating peripheral microvessel permeability and BBB function, the particular contribution of Epac proteins in this context has received little attention (Bader et al. 2017).

Phosphodiesterases' inhibitors

PDEs are enzymes that hydrolyze the second messengers cAMP and/or cGMP throughout the brain, playing a significant role in cell signaling (Beavo 1995). Different types of cognitive deficiencies and improper growth or degeneration in various brain areas are characteristics of specific brain illnesses. Therefore, different PDEs may be targeted for therapeutic advantages for various disorders, and it is

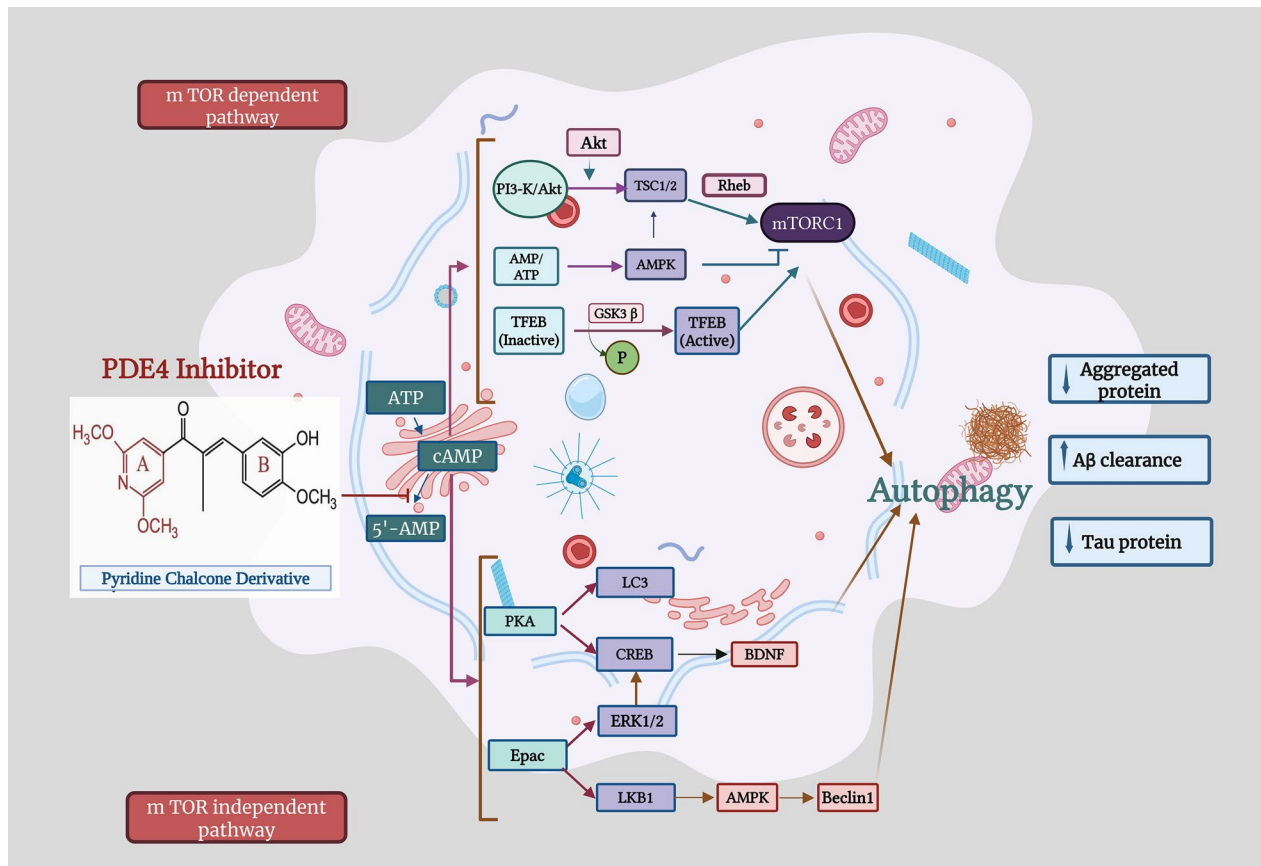


Fig. 4 Autophagy-linked signaling mechanisms in Alzheimer's disease. Both mTOR-dependent and mTOR-independent pathways are highlighted in the figure, which illustrates autophagy-linked signaling pathways in Alzheimer's disease. Although autophagy is inhibited by the mTOR-dependent system, it can be altered to encourage the breakdown of harmful protein aggregates in AD. Autophagy is

activated by the mTOR-independent pathway, which is controlled by AMPK and Beclin-1 and offers an alternate avenue for cellular clearance. The buildup of tau and amyloid-beta is exacerbated by the dysregulation of both processes. Potential treatment approaches to restore autophagic equilibrium in Alzheimer's disease include focusing on these pathways

crucial for drug development to match the proper PDE target with each CNS ailment. The largest of the 11 PDE families, the PDE4 family, will be the one to be specifically targeted in this review to improve cognition (Menniti et al. 2006).

Use of PDE4 in the treatment of Alzheimer's disease through the cAMP pathway

For CNS diseases, PDE4 has emerged as a potential therapeutic target due to its crucial role in modulating cAMP signaling, which is implicated in substance dependence (Xiao et al. 2014). The goal of developing PDE4B-targeting medications for Alzheimer's disease (AD) is to minimize side effects, especially emesis, which has been linked to non-selective PDE4 inhibition, while still achieving therapeutic success (J. Chen et al. 2025). Inhibiting PDE4 has shown potential as a pharmacological target for treating cognitive deficits, as seen in research where the PDE4 antagonist roflumilast enhanced memory in rats at non-emetic dosages

(Vanmierlo et al. 2016). In vivo experiments employing transgenic AD mouse models, where cAMP-elevating drugs like forskolin have been demonstrated to promote autophagy and diminish amyloid-beta buildup, are one experimental method to validate the function of the cAMP pathway in autophagy regulation for AD therapy. Its function in autophagy-mediated neuroprotection may be further clarified by genetic modification research employing CRISPR to upregulate PKA, a crucial cAMP effector (Button et al. 2015; Sumitomo and Tomoda 2021).

Roflumilast has been shown in preclinical research to improve memory and learning while also reducing depressive behaviors among APP/PS1 transgenic mice. This effect is most likely caused via PDE4B/PDE4D-mediated cAMP/CREB/BDNF signaling. As a result, roflumilast may be an effective AD therapy, particularly when memory loss and depression are present (H. Wang et al. 2020a, b). PDE4D-NAMs (negative allosteric modulators) have demonstrated cognitive advantages by enhancing communication by

activating the cAMP-response element-binding protein (CREB)/cAMP/protein kinase A pathway for consolidation of memory, whereas PDE4B-NAMs have had the ability to decrease neurological inflammation by inhibiting microglia cytokine production brought on by brain amyloid induction (Gurney et al. 2015). In animal models of CNS disorders, PDE4 inhibition has demonstrated the ability to correct functional deficits and restore compromised cAMP functioning, with neuroplasticity and anti-inflammatory effects being the primary mechanisms of action for PDE4 inhibitors (Blokland et al. 2019a, b). In the setting of APP-PS1 mice, decreased MAPK14 activity used as a treatment strategy to address physiological deficiencies inside neurons in the brains of people with AD. MAPK14 has been connected to autophagic activities (Alam and Scheper 2016). The novel PDE4D inhibitor GEBR-32a has been shown to work can repair the hippocampus long-term potentiation deficit and enhance memory in AD transgenic mice, making it a viable cognitive-enhancing medication and treatment for AD (Ricciarelli et al. 2017). Selective PDE4 inhibitors hold promise as effective medications for treating metabolic diseases associated with aging and offer a potential strategy for managing age-related metabolic dysfunctions (Fernández-Araujo et al. 2015). After receiving Yessotoxins (YTX) for 48 h, PDE4A-dependent autophagy, a non-apoptotic form of planned cell death, is induced (Blokland et al. 2019a, b).

In a study using mice with aggregated A β 42, long-form PDE4D silencing reversed cAMP depletion and memory impairments in the brain while altering critical memory signaling molecules (Saletu et al. 1992). In a research study including individuals with minor to serious AD, denbufylline, a PDE4 inhibitor, significantly improved various

cognitive scores compared to placebo (Sebastiani et al. 2006). PDE4B activation appears to be directly related to the commencement of TNF- release by the same cells, as indicated by the fact that inhibiting PDE4B activity reduces TNF-synthesis and release. These findings raise the possibility of using PDE4B inhibitors as anti-inflammatory medicines to treat AD. The release of TNF- from microglial cells induced by sA β is significantly decreased by blocking PDE4B, reaching up to a 70% reduction. This emphasizes the importance of cAMP as a key mediator in A β -induced microglial activation (Van Duinen et al. 2018). Clinical research has investigated the use of donepezil, a cholinesterase inhibitor frequently used to treat AD, in conjunction with the PDE4 inhibitor roflumilast. PDE4 inhibitors may increase the cognitive advantages of current treatments, according to preliminary findings that showed improvements in verbal learning performance (Abdallah 2024).

Autophagy is essential for removing damaged organelles and misfolded proteins, two functions that are compromised in AD. It has been demonstrated that curcumin, for instance, inhibits amyloid- β aggregation and increases autophagic flow by modulating the mTOR pathway and inhibiting GSK-3 β . By blocking inositol (Deng et al. 2022). Table 2 summarizes the current status of PDE4 inhibitors in clinical trials, highlighting their potential applications and ongoing research efforts.

Limitations of PDE4 inhibitors

The PDE4 family is the largest and most diversely expressed phosphodiesterase (PDE) family in the human body, making it one of the most complex as well. This diversity in

Table 2 Current status of PDE4 inhibitors in clinical trials

Compounds	Dosing, treatment	Subjects Status	Status	Result	NCT Number
Roflumilast	50 and 100 μ g, single oral dose administration	Patients With (Amnesic) Mild Cognitive Impairment (MCI) or Mild Dementia	Phase II	Not yet available	NCT04658654 (2021)
Roflumilast	100, 300 and 1000 μ g, single oral dose administration	Healthy young adult	Phase II	Roflumilast has potential memory enhancer in humans at low-dose	NCT01433666 (2011), (Fernández-Araujo et al. 2015)
Roflumilast	Dose was not disclosed, single administration with or without donepezil (10 mg)	Healthy young adult	Phase I	Not yet available	NCT02051335 (2014)
Roflumilast HCl (Daliresp or Daxas)	100, 250, and 1000 μ g, single oral dose administration	Healthy elderly patients	Phase II	PDE4 inhibitor roflumilast improves verbal memory performance in old participants	(Blokland, Van Duinen, et al., 2019)
MK0952	Dose was not disclosed, oral dose administration	Patients With Mild-to-Moderate AD	Phase II	Not yet available	NCT00362024 (2006)
BPN14770 (Zatomilast)	Dose was not disclosed, oral dose administration	Healthy adult	Phase I	Not yet available	NCT02648672 (2015)

expression means that inhibiting PDE4 can lead to a wide range of physiological effects, presenting opportunities for drug development across multiple therapeutic areas, including memory and cognitive enhancement. PDE4 enzymes, play a crucial role in hydrolyzing cyclic adenosine monophosphate (cAMP) in various mammalian cells and tissues, which is vital for numerous cellular signaling pathways. The necessity of precisely regulating autophagic activity to preserve neuronal health is highlighted by the fact that excessive autophagy activation in neurons can be harmful and can cause cell death. Targeting important regulatory molecules and pathways, including mTOR inhibitors or AMPK activators, is one strategy to manage autophagy and attain a balanced autophagic response (Button et al. 2015; Sumitomo and Tomoda 2021). There are numerous important obstacles to overcome before autophagy-targeting treatments may be transferred from animal models to human clinical trials. Since animal models sometimes fall short of adequately simulating the intricacies of human disease, physiological and genetic variances between species can result in differences in therapeutic efficacy and safety. These constraints are highlighted by the fact that, on average, less than 8% of translations from animal models to clinical cancer studies are effective (Mak et al. 2014).

Research into PAN-PDE4 inhibitors has shown promise for treating conditions, such as depression, schizophrenia, and neurodegenerative diseases, owing to their potential to enhance cognitive function. However, the most common side effects associated with these inhibitors, particularly gastrointestinal reactions, include nausea, vomiting, and diarrhea. These adverse effects were first noted with the prototype PDE4 inhibitor, Rolipram, and are observed to varying degrees across the entire class of PAN-PDE4 inhibitors (Jin et al. 2006).

Additionally, ongoing research is exploring the potential of selectively targeting specific PDE4 isoforms to minimize side effects while maximizing therapeutic benefits. Such advancements may improve patient adherence to treatment and expand the applicability of PDE4 inhibitors in clinical practice. Understanding the intricate roles of different PDE4 isoforms will be crucial for developing more effective therapies with fewer side effects, particularly for conditions involving cognitive decline (Heaslip and Evans 1995; Hebenstreit et al. 1989).

Conclusion

According to the authors, healthy neuronal aging depends on a functional autophagy system, despite the fact that the majority of the information in the review comes from research using animal models. AD and other neurodegenerative illnesses are significantly influenced by disrupted

neuronal autophagy. The regulation of the mTOR pathway, the degradation of the A β and tau proteins, the activation of AMPK, the upregulation of TFEB, and the cAMP pathway are all processes in which autophagy plays a critical role and may all have an effect on AD. Considering these findings, therapeutic approaches targeting autophagy with the cAMP pathway hold promise in the exploration of innovative strategies for managing AD using PDE4B drugs. By addressing autophagy-related mechanisms, new avenues for treatment in this neurodegenerative disease could be unveiled.

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Declarations

Conflict of interest All authors declare that there is no conflict of interest.

References

- Abdallah AE (2024) Review on anti-alzheimer drug development: approaches, challenges and perspectives. *RSC Adv* 14(16):11057–11088. <https://doi.org/10.1039/d3ra08333k>
- Abusnina A, Lugnier C (2017) Therapeutic potentials of natural compounds acting on cyclic nucleotide phosphodiesterase families. *Cell Signal* 39:55–65. <https://doi.org/10.1016/j.cellsig.2017.07.018>
- Agholme L, Hallbeck M, Benedikz E, Marcusson J, Kågedal K (2012) Amyloid- β secretion, generation, and lysosomal sequestration in response to proteasome inhibition: involvement of autophagy. *J Alzheimer's Dis* 31(2):343–358. <https://doi.org/10.3233/JAD-2012-120001>
- Al-Harbi NO, Imam F, Al-Harbi MM, Qamar W, Algerian K, Anwer MK, Alharbi M, Almudimeegh S, Alhamed AS, Alshamrani AA (2023) Effect of Apremilast on LPS-induced immunomodulation and inflammation via activation of Nrf2/HO-1 pathways in rat lungs. *Saudi Pharmaceut J* 31(7):1327–1338. <https://doi.org/10.1016/j.jsps.2023.07.016>
- Alam J, Scheper W (2016) Targeting neuronal MAPK14/p38 α activity to modulate autophagy in the Alzheimer disease brain.

- Autophagy 12(12):2516–2520. <https://doi.org/10.1080/15548627.2016.1238555>
- An W-L, Cowburn RF, Li L, Braak H, Alafuzoff I, Iqbal K, Iqbal I-G, Winblad B, Pei J-J (2003) Up-Regulation of phosphorylated/activated p70 S6 kinase and its relationship to neurofibrillary pathology in alzheimer's disease. *Am J Pathol* 163(2):591–607. [https://doi.org/10.1016/S0002-9440\(10\)63687-5](https://doi.org/10.1016/S0002-9440(10)63687-5)
- Bader A, Bintig W, Begandt D, Klett A, Siller IG, Gregor C, Schaarschmidt F, Weksler B, Romero I, Couraud P, Hell SW, Ngezahayo A (2017) Adenosine receptors regulate gap junction coupling of the human cerebral microvascular endothelial cells hCMEC/D3 by Ca^{2+} influx through cyclic nucleotide-gated channels. *J Physiol* 595(8):2497–2517. <https://doi.org/10.1113/JP273150>
- Barmaki H, Nourazarian A, Khaki-Khatibi F (2023) Proteostasis and neurodegeneration: a closer look at autophagy in Alzheimer's disease. *Front Aging Neurosci* 15:1281338. <https://doi.org/10.3389/fnagi.2023.1281338>
- Beavo JA (1995) Cyclic nucleotide phosphodiesterases: functional implications of multiple isoforms. *Physiol Rev* 75(4):725–748. <https://doi.org/10.1152/physrev.1995.75.4.725>
- Di Benedetto G, Iannucci LF, Surdo NC, Zanin S, Conca F, Grisan F, Gerbino A, Lefkimiatis K (2021) Compartmentalized signaling in aging and neurodegeneration. *Cells* 10(2):464. <https://doi.org/10.3390/cells10020464>
- Berger Z, Ravikumar B, Menzies FM, Oroz LG, Underwood BR, Pangalos MN, Schmitt I, Wullner U, Evert BO, O'Kane CJ, Rubinsztein DC (2006) Rapamycin alleviates toxicity of different aggregate-prone proteins. *Hum Mol Genet* 15(3):433–442. <https://doi.org/10.1093/hmg/ddi458>
- Bettens K, Slegers K, Van Broeckhoven C (2013) Genetic insights in Alzheimer's disease. *The Lancet Neurol* 12(1):92–104. [https://doi.org/10.1016/S1474-4422\(12\)70259-4](https://doi.org/10.1016/S1474-4422(12)70259-4)
- Blokland A, Van Duinen MA, Sambeth A, Heckman PRA, Tsai M, Lahu G, Uz T, Prickaerts J (2019b) Acute treatment with the PDE4 inhibitor roflumilast improves verbal word memory in healthy old individuals: a double-blind placebo-controlled study. *Neurobiol Aging* 77:37–43. <https://doi.org/10.1016/j.neurobiolaging.2019.01.014>
- Blokland A, Heckman P, Vanmierlo T, Schreiber R, Paes D, Prickaerts J (2019a) Phosphodiesterase type 4 inhibition in CNS diseases. *Trends Pharmacol Sci* 40(12):971–985. <https://doi.org/10.1016/j.tips.2019.10.006>
- Boland B, Kumar A, Lee S-H, Platt FM, Wegiel J, Yu W-H, Nixon RA (2008) Autophagy induction and autophagosome clearance in neurons: relationship to autophagic pathology in Alzheimer's disease. *J Neurosci* 28(27):6926–6937. <https://doi.org/10.1523/JNEUROSCI.0800-08.2008>
- Boland B, Smith DA, Mooney D, Jung SS, Walsh DM, Platt FM (2010) Macroautophagy is not directly involved in the metabolism of amyloid precursor protein. *J Biol Chem* 285(48):37415–37426. <https://doi.org/10.1074/jbc.M110.186411>
- Boswell-Smith V, Spina D, Page CP (2006) Phosphodiesterase inhibitors. *Br J Pharmacol* 147(S1):S252–S257. <https://doi.org/10.1038/sj.bjp.0706495>
- Bové J, Martínez-Vicente M, Vila M (2011) Fighting neurodegeneration with rapamycin: mechanistic insights. *Nat Rev Neurosci* 12(8):437–452. <https://doi.org/10.1038/nrn3068>
- Breijyeh Z, Karaman R (2020) Comprehensive review on Alzheimer's disease: causes and treatment. *Molecules* 25(24):5789
- Brown MR, Bondada V, Keller JN, Thorpe J, Geddes JW (2005) Proteasome or calpain inhibition does not alter cellular tau levels in neuroblastoma cells or primary neurons. *J Alzheimer's Dis* 7(1):15–24
- Button RW, Luo S, Rubinsztein DC (2015) Autophagic activity in neuronal cell death. *Neurosci Bull* 31(4):382–394. <https://doi.org/10.1007/s12264-015-1528-y>
- Caccamo A, Magri A, Medina DX, Wisely EV, López-Aranda MF, Silva AJ, Oddo S (2013) mTOR regulates tau phosphorylation and degradation: implications for Alzheimer's disease and other tauopathies. *Aging Cell* 12(3):370–380. <https://doi.org/10.1111/ace.12057>
- Caccamo A, Maldonado MA, Majumder S, Medina DX, Holbein W, Magri A, Oddo S (2011) Naturally secreted amyloid- β increases mammalian target of rapamycin (mTOR) activity via a PRAS40-mediated mechanism. *J Biol Chem* 286(11):8924–8932. <https://doi.org/10.1074/jbc.M110.180638>
- Calabrese MF, Rajamohan F, Harris MS, Caspers NL, Magyar R, Withka JM, Wang H, Borzilleri KA, Sahasrabudhe PV, Hoth LR, Geoghegan KF, Han S, Brown J, Subashi TA, Reyes AR, Frisbie RK, Ward J, Miller RA, Landro JA, Kurumbail RG (2014) Structural basis for AMPK activation: natural and synthetic ligands regulate kinase activity from opposite poles by different molecular mechanisms. *Structure* 22(8):1161–1172. <https://doi.org/10.1016/j.str.2014.06.009>
- Caselli RJ, Beach TG, Knopman DS, Graff-Radford NR (2017) Alzheimer disease. *Mayo Clin Proc* 92(6):978–994. <https://doi.org/10.1016/j.mayocp.2017.02.011>
- Chandra S, Jana M, Pahan K (2018) Aspirin induces lysosomal biogenesis and attenuates amyloid plaque pathology in a mouse model of Alzheimer's disease via PPAR α . *J Neurosci* 38(30):6682–6699. <https://doi.org/10.1523/JNEUROSCI.0054-18.2018>
- Chen Y, Huang X, Zhang Y, Rockenstein E, Bu G, Golde TE, Masliah E, Xu H (2012) Alzheimer's β -Secretase (BACE1) regulates the cAMP/PKA/CREB pathway independently of β -Amyloid. *J Neurosci* 32(33):11390–11395. <https://doi.org/10.1523/JNEUROSCI.0757-12.2012>
- Chen J, Zhu Z, Xu F, Dou B, Sheng Z, Xu Y (2025) Phosphodiesterase 4 inhibition in neuropsychiatric disorders associated with Alzheimer's disease. *Cells* 14(3):164. <https://doi.org/10.3390/cells14030164>
- Cohen TJ, Guo JL, Hurtado DE, Kwong LK, Mills IP, Trojanowski JQ, Lee VMY (2011) The acetylation of tau inhibits its function and promotes pathological tau aggregation. *Nat Commun* 2(1):252. <https://doi.org/10.1038/ncomms1255>
- Cook C, Stankowski JN, Carlomagno Y, Stetler C, Petrucelli L (2014) Acetylation: a new key to unlock tau's role in neurodegeneration. *Alzheimer's Res Ther* 6(3):29. <https://doi.org/10.1186/alzrt259>
- DeTure MA, Dickson DW (2019) The neuropathological diagnosis of Alzheimer's disease. *Mol Neurodegener* 14(1):32. <https://doi.org/10.1186/s13024-019-0333-5>
- DeVos SL, Corjuc BT, Oakley DH, Nobuhara CK, Bannion RN, Chase A, Commins C, Gonzalez JA, Dooley PM, Frosch MP, Hyman BT (2018) Synaptic tau seeding precedes tau pathology in human Alzheimer's disease brain. *Front Neurosci* 12:267. <https://doi.org/10.3389/fnins.2018.00267>
- Deng Z, Dong Y, Zhou X, Lu J-H, Yue Z (2022) Pharmacological modulation of autophagy for Alzheimer's disease therapy: opportunities and obstacles. *Acta Pharmaceutica Sinica B* 12(4):1688–1706. <https://doi.org/10.1016/j.apsb.2021.12.009>
- Ditaranto K, Tekirian TL, Yang AJ (2001) Lysosomal membrane damage in soluble A β -Mediated cell death in Alzheimer's disease. *Neurobiol Dis* 8(1):19–31. <https://doi.org/10.1006/nbdi.2000.0364>
- Divya D, Thekkekkara D, Sai TE, Ramakrishna D, Manjula SN, Goswami SK (2025) Molecular etiopathogenesis of neurodegenerative diseases. In *The Neurodegeneration Revolution* (pp. 17–72). Elsevier. <https://doi.org/10.1016/B978-0-443-28822-7.00002-7>
- Du H, Guo L, Wu X, Sosunov AA, McKhann GM, Chen JX, Yan SS (2014) Cyclophilin D deficiency rescues A β -impaired PKA/CREB signaling and alleviates synaptic degeneration. *Biochimica et Biophysica Acta (BBA) – Mol Basis Dis* 1842(12):2517–2527. <https://doi.org/10.1016/j.bbadis.2013.03.004>

- Van Duinen MA, Sambeth A, Heckman PRA, Smit S, Tsai M, Lahu G, Uz T, Blokland A, Prickaerts J (2018) Acute administration of roflumilast enhances immediate recall of verbal word memory in healthy young adults. *Neuropharmacology* 131:31–38. <https://doi.org/10.1016/j.neuropharm.2017.12.019>
- Fernández-Araujo A, Alfonso A, Vieytes MR, Botana LM (2015) Key role of phosphodiesterase 4A (PDE4A) in autophagy triggered by yessotoxin. *Toxicology* 329:60–72
- Franco-Juárez B, Coronel-Cruz C, Hernández-Ochoa B, Gómez-Manzo S, Cárdenas-Rodríguez N, Arreguin-Espinosa R, Bandala C, Canseco-Ávila LM, Ortega-Cuellar D (2022) TFEB; beyond its role as an autophagy and lysosomes regulator. *Cells* 11(19):3153. <https://doi.org/10.3390/cells11193153>
- Frederick C, Ando K, Leroy K, Héraud C, Suain V, Buée L, Brion J-P (2015) Rapamycin ester analog CCI-779/Temsirolimus alleviates tau pathology and improves motor deficit in mutant tau transgenic mice. *J Alzheimer's Dis* 44(4):1145–1156. <https://doi.org/10.3233/JAD-142097>
- Canley IG, Lam DH, Wang J, Ding X, Chen S, Jiang X (2009) ULK1-ATG13-FIP200 complex mediates mTOR signaling and is essential for autophagy. *J Biol Chem* 284(18):12297–12305. <https://doi.org/10.1074/jbc.M900786200>
- Ge Y, Zhou M, Chen C, Wu X, Wang X (2022) Role of AMPK mediated pathways in autophagy and aging. *Biochimie* 195:100–113. <https://doi.org/10.1016/j.biochi.2021.11.008>
- Gomaa AA, Farghaly HSM, Ahmed AM, Hemida FK (2023) Intermittent treatment with Apremilast, a phosphodiesterase-4 inhibitor, ameliorates Alzheimer's-like pathology and symptoms through multiple targeting actions in aged T2D rats. *Int Immunopharmacol* 117:109927. <https://doi.org/10.1016/j.intimp.2023.109927>
- Guo W, Qian L, Zhang J, Zhang W, Morrison A, Hayes P, Wilson S, Chen T, Zhao J (2011) Sirt1 overexpression in neurons promotes neurite outgrowth and cell survival through inhibition of the mTOR signaling. *J Neurosci Res* 89(11):1723–1736. <https://doi.org/10.1002/jnr.22725>
- Gurney ME, D'Amato EC, Burgin AB (2015) Phosphodiesterase-4 (PDE4) molecular pharmacology and Alzheimer's disease. *Neurotherapeutics* 12:49–56
- Halls ML, Cooper DMF (2017) Adenylyl cyclase signaling complexes – pharmacological challenges and opportunities. *Pharmacol Ther* 172:171–180. <https://doi.org/10.1016/j.pharmthera.2017.01.001>
- Hara T, Nakamura K, Matsui M, Yamamoto A, Nakahara Y, Suzuki-Migishima R, Yokoyama M, Mishima K, Saito I, Okano H, Mizushima N (2006) Suppression of basal autophagy in neural cells causes neurodegenerative disease in mice. *Nature* 441(7095):885–889. <https://doi.org/10.1038/nature04724>
- Hatzelmann A, Morcillo EJ, Lungarella G, Adnot S, Sanjar S, Beume R, Schudt C, Tenor H (2010) The preclinical pharmacology of roflumilast—a selective, oral phosphodiesterase 4 inhibitor in development for chronic obstructive pulmonary disease. *Pulm Pharmacol Ther* 23(4):235–256. <https://doi.org/10.1016/j.pupt.2010.03.011>
- Heaslip RJ, Evans DY (1995) Emetic, central nervous system and pulmonary activities of rolipram in the dog. *Eur J Pharmacol* 286(3):281–290. [https://doi.org/10.1016/0014-2999\(95\)00457-2](https://doi.org/10.1016/0014-2999(95)00457-2)
- Hebenstreit G, Fellerer K, Fichte K, Fischer G, Geyer N, Meya U, Sastre-y-Hernández M, Schöny W, Schratzer M, Soukup W, Trampitsch E, Varosanec S, Zawada E, Zöchling R (1989) Rolipram in major depressive disorder: results of a double-blind comparative study with imipramine. *Pharmacopsychiatry* 22(04):156–160. <https://doi.org/10.1055/s-2007-1014599>
- Heckmann BL, Teubner BJW, Tummers B, Boada-Romero E, Harris L, Yang M, Guy CS, Zakharenko SS, Green DR (2019) LC3-Associated endocytosis facilitates β -Amyloid clearance and mitigates neurodegeneration in murine Alzheimer's disease. *Cell* 178(3):536–551.e14. <https://doi.org/10.1016/j.cell.2019.05.056>
- Holmes BB, Furman JL, Mahan TE, Yamasaki TR, Mirbaha H, Eades WC, Belaygorod L, Cairns NJ, Holtzman DM, Diamond MI (2014) Proteopathic tau seeding predicts tauopathy in vivo. *Proc Natl Acad Sci* 111(41):E4376–E4385. <https://doi.org/10.1073/pnas.1411649111>
- Holtzman DM, Morris JC, Goate AM (2011) Alzheimer's disease: the challenge of the second century. *Sci Transl Med*. <https://doi.org/10.1126/scitranslmed.3002369>
- Inoki K, Ouyang H, Zhu T, Lindvall C, Wang Y, Zhang X, Yang Q, Bennett C, Harada Y, Stankunas K, Wang C, He X, MacDougald OA, You M, Williams BO, Guan K-L (2006) TSC2 integrates Wnt and energy signals via a coordinated phosphorylation by AMPK and GSK3 to regulate cell growth. *Cell* 126(5):955–968. <https://doi.org/10.1016/j.cell.2006.06.055>
- Jiang T, Yu J-T, Zhu X-C, Tan M-S, Wang H-F, Cao L, Zhang Q-Q, Shi J-Q, Gao L, Qin H, Zhang Y-D, Tan L (2014) Temsirolimus promotes autophagic clearance of amyloid- β and provides protective effects in cellular and animal models of Alzheimer's disease. *Pharmacol Res* 81:54–63. <https://doi.org/10.1016/j.phrs.2014.02.008>
- Jin SL, Richter W, Conti M (2006) Insights into the physiological functions of PDE4 from knockout mice. In *Cyclic Nucleotide Phosphodiesterases in Health and Disease* (pp. 323–346)
- Johnson CW, Melia TJ, Yamamoto A (2012) Modulating macroautophagy: a neuronal perspective. *Future Med Chem* 4(13):1715–1731. <https://doi.org/10.4155/fmc.12.112>
- Ju T-C, Chen H-M, Lin J-T, Chang C-P, Chang W-C, Kang J-J, Sun C-P, Tao M-H, Tu P-H, Chang C, Dickson DW, Chern Y (2011) Nuclear translocation of AMPK- α 1 potentiates striatal neurodegeneration in Huntington's disease. *J Cell Biol* 194(2):209–227. <https://doi.org/10.1083/jcb.201105010>
- Kang S, Son SM, Baik SH, Yang J, Mook-Jung I (2019) Autophagy-mediated secretory pathway is secretory pathway for both normal and pathological tau in neurons. *J Alzheimer's Dis* 70(3):667–680. <https://doi.org/10.3233/JAD-190180>
- Kelly MP (2018) Cyclic nucleotide signaling changes associated with normal aging and age-related diseases of the brain. *Cell Signal* 42:281–291
- Kesidou E, Lagoudaki R, Touloumi O, Poulatsidou KN, Simeonidou C (2013) Autophagy and neurodegenerative disorders. *Neural Regen Res* 8:2275–2283. <https://doi.org/10.3969/j.issn.1673-5374.2013.24.001>
- Klionsky DJ, Clegg JM, Dunn WA, Emr SD, Sakai Y, Sandoval IV, Sibirny A, Subramani S, Thumm M, Veenhuis M, Ohsumi Y (2003) A unified nomenclature for yeast autophagy-related genes. *Dev Cell* 5(4):539–545. [https://doi.org/10.1016/S1534-5807\(03\)00296-X](https://doi.org/10.1016/S1534-5807(03)00296-X)
- Kobayashi M, Kubo S, Iwata M, Ohtsu Y, Takahashi K, Shimizu Y (2011) ASP3258, an orally active potent phosphodiesterase 4 inhibitor with low emetic activity. *Int Immunopharmacol* 11(6):732–739. <https://doi.org/10.1016/j.intimp.2011.01.023>
- Komatsu M, Waguri S, Chiba T, Murata S, Iwata J, Tanida I, Ueno T, Koike M, Uchiyama Y, Kominami E, Tanaka K (2006) Loss of autophagy in the central nervous system causes neurodegeneration in mice. *Nature* 441(7095):880–884. <https://doi.org/10.1038/nature04723>
- Kondaveeti SN, Thummuru ES, Thekkekkara D, Narayanan TL, Tausif YM, Srivalli Ch MR, Babu A (2024) Therapeutic potential of Simvastatin and Doxycycline combination for Alzheimer's disease. *J Appl Pharm Sci*. <https://doi.org/10.7324/JAPS.2024.167710>
- Kurumbail RG, Calabrese MF (2016). Structure and Regulation of AMPK. In *AMP-activated Protein Kinase* (pp. 3–22). https://doi.org/10.1007/978-3-319-43589-3_1
- Lambert J-C, Ibrahim-Verbaas CA, Harold D, Naj AC, Sims R, Bellenguez C, Jun G, DeStefano AL, Bis JC, Beecham GW,

- Grenier-Boley B, Russo G, Thornton-Wells TA, Jones N, Smith AV, Chouraki V, Thomas C, Ikram MA, Zelenika D, Amouyel P (2013) Meta-analysis of 74,046 individuals identifies 11 new susceptibility loci for Alzheimer's disease. *Nat Genet* 45(12):1452–1458. <https://doi.org/10.1038/ng.2802>
- Lee J-A (2012) Neuronal autophagy: a housekeeper or a fighter in neuronal cell survival? *Exp Neurobiol* 21(1):1–8. <https://doi.org/10.5607/en.2012.21.1.1>
- Lee MJ, Lee JH, Rubinsztein DC (2013) Tau degradation: the ubiquitin–proteasome system versus the autophagy-lysosome system. *Prog Neurobiol* 105:49–59. <https://doi.org/10.1016/j.pneurobio.2013.03.001>
- Li X, Alafuzoff I, Soininen H, Winblad B, Pei J (2005) Levels of mTOR and its downstream targets 4E-BP1, eEF2, and eEF2 kinase in relationships with tau in Alzheimer's disease brain. *FEBS J* 272(16):4211–4220. <https://doi.org/10.1111/j.1742-4658.2005.04833.x>
- Li Y, Xu M, Ding X, Yan C, Song Z, Chen L, Huang X, Wang X, Jian Y, Tang G, Tang C, Di Y, Mu S, Liu X, Liu K, Li T, Wang Y, Miao L, Guo W, Yang C (2016) Protein kinase C controls lysosome biogenesis independently of mTORC1. *Nat Cell Biol* 18(10):1065–1077. <https://doi.org/10.1038/ncb3407>
- Lipton JO, Sahin M (2014) The neurology of mTOR. *Neuron* 84(2):275–291. <https://doi.org/10.1016/j.neuron.2014.09.034>
- Loewith R, Jacinto E, Wullschlegel S, Lorberg A, Crespo JL, Bonenfant D, Oppliger W, Jenoe P, Hall MN (2002) Two TOR complexes, only one of which is rapamycin sensitive, have distinct roles in cell growth control. *Mol Cell* 10(3):457–468. [https://doi.org/10.1016/S1097-2765\(02\)00636-6](https://doi.org/10.1016/S1097-2765(02)00636-6)
- Lopez Salon M, Pasquini L, Besio Moreno M, Pasquini JM, Soto E (2003) Relationship between β -amyloid degradation and the 26S proteasome in neural cells. *Exp Neurol* 180(2):131–143. [https://doi.org/10.1016/S0014-4886\(02\)00060-2](https://doi.org/10.1016/S0014-4886(02)00060-2)
- Majd S, Power JHT (2018) Oxidative stress and decreased mitochondrial superoxide dismutase 2 and peroxiredoxins 1 and 4 based mechanism of concurrent activation of AMPK and mTOR in Alzheimer's disease. *Curr Alzheimer Res* 15(8):764–776. <https://doi.org/10.2174/1567205015666180223093020>
- Mak IW, Evaniew N, Ghert M (2014) Lost in translation: animal models and clinical trials in cancer treatment. *Am J Transl Res* 6(2):114–118
- Martín D, Salinas M, López-Valdaliso R, Serrano E, Recuero M, Cuadrado A (2001) Effect of the Alzheimer amyloid fragment A β (25–35) on Akt/PKB kinase and survival of PC12 cells. *J Neurochem* 78(5):1000–1008. <https://doi.org/10.1046/j.1471-4159.2001.00472.x>
- Masters CL, Bateman R, Blennow K, Rowe CC, Sperling RA, Cummings JL (2015) Alzheimer's disease. *Nat Rev Dis Primers* 1(1):15056. <https://doi.org/10.1038/nrdp.2015.56>
- McGowan E, Pickford F, Kim J, Onstead L, Eriksen J, Yu C, Skipper L, Murphy MP, Beard J, Das P, Jansen K, DeLucia M, Lin W-L, Dolios G, Wang R, Eckman CB, Dickson DW, Hutton M, Hardy J, Golde T (2005) A β 42 is essential for parenchymal and vascular amyloid deposition in mice. *Neuron* 47(2):191–199. <https://doi.org/10.1016/j.neuron.2005.06.030>
- Meijer AJ, Codogno P (2006) Signalling and autophagy regulation in health, aging and disease. *Mol Aspects Med* 27(5–6):411–425. <https://doi.org/10.1016/j.mam.2006.08.002>
- Meijer AJ, Lorin S, Blommaert EF, Codogno P (2015) Regulation of autophagy by amino acids and MTOR-dependent signal transduction. *Amino Acids* 47(10):2037–2063. <https://doi.org/10.1007/s00726-014-1765-4>
- Menniti FS, Faraci WS, Schmidt CJ (2006) Phosphodiesterases in the CNS: targets for drug development. *Nat Rev Drug Discov* 5(8):660–670. <https://doi.org/10.1038/nrd2058>
- Metaxakis A, Ploumi C, Tavernarakis N (2018) Autophagy in age-associated neurodegeneration. *Cells* 7(5):37. <https://doi.org/10.3390/cells7050037>
- Miners JS, Clarke P, Love S (2017) Clusterin levels are increased in Alzheimer's disease and influence the regional distribution of A β . *Brain Pathol (Zurich, Switzerland)* 27(3):305–313. <https://doi.org/10.1111/bpa.12392>
- Nabar NR, Kehrl JH (2017) The transcription factor EB links cellular stress to the immune response. *The Yale J Biol Med* 90(2):301–315
- Nangare S, Patil P (2022) Prevalence, distribution, treatment, and modern methods for in vitro diagnosis of Alzheimer's disease in India: Challenges and future prospective. *Thai J Pharm Sci* 46(2):149
- Napolitano G, Esposito A, Choi H, Matarese M, Benedetti V, Di Malta C, Monfregola J, Medina DL, Lippincott-Schwartz J, Ballabio A (2018) mTOR-dependent phosphorylation controls TFEB nuclear export. *Nat Commun* 9(1):3312. <https://doi.org/10.1038/s41467-018-05862-6>
- Narayanan L, Thekkekkara D, Kondaveeti SN, Babu A, Sailen BA, Chidambaram SB, Manjula SN (2024) Deciphering the intricate role of mTOR signaling and autophagy in Parkinson's disease and therapeutic prospects. *J Appl Pharm Sci*. <https://doi.org/10.7324/JAPS.2023.156692>
- Nilsson P, Loganathan K, Sekiguchi M, Matsuba Y, Hui K, Tsubuki S, Tanaka M, Iwata N, Saito T, Saido TC (2013) A β secretion and plaque formation depend on autophagy. *Cell Rep* 5(1):61–69. <https://doi.org/10.1016/j.celrep.2013.08.042>
- Nixon RA (2013) The role of autophagy in neurodegenerative disease. *Nat Med* 19(8):983–997. <https://doi.org/10.1038/nm.3232>
- Palmieri M, Pal R, Nelvagal HR, Lotfi P, Stinnett GR, Seymour ML, Chaudhury A, Bajaj L, Bondar VV, Bremner L, Saleem U, Tse DY, Sanagasetti D, Wu SM, Neilson JR, Pereira FA, Pautler RG, Rodney GG, Cooper JD, Sardiello M (2017) mTORC1-independent TFEB activation via Akt inhibition promotes cellular clearance in neurodegenerative storage diseases. *Nat Commun* 8(1):14338. <https://doi.org/10.1038/ncomms14338>
- Parr C, Carzaniga R, Gentleman SM, Van Leuven F, Walter J, Sastre M (2012) Glycogen synthase kinase 3 inhibition promotes lysosomal biogenesis and autophagic degradation of the Amyloid- β precursor protein. *Mol Cell Biol* 32(21):4410–4418. <https://doi.org/10.1128/MCB.00930-12>
- Perluigi M, Di Domenico F, Butterfield DA (2015) mTOR signaling in aging and neurodegeneration: at the crossroad between metabolism dysfunction and impairment of autophagy. *Neurobiol Dis* 84:39–49. <https://doi.org/10.1016/j.nbd.2015.03.014>
- Pickford F, Masliah E, Britschgi M, Lucin K, Narasimhan R, Jaeger PA, Small S, Spencer B, Rockenstein E, Levine B, Wyss-Coray T (2008) The autophagy-related protein beclin 1 shows reduced expression in early Alzheimer disease and regulates amyloid β accumulation in mice. *J Clin Invest* 118:2190–2199. <https://doi.org/10.1172/JCI33585>
- Piras A, Collin L, Grüniger F, Graff C, Rönnebeck A (2016) Autophagic and lysosomal defects in human tauopathies: analysis of post-mortem brain from patients with familial Alzheimer disease, corticobasal degeneration and progressive supranuclear palsy. *Acta Neuropathol Commun* 4(1):22. <https://doi.org/10.1186/s40478-016-0292-9>
- Polito VA, Li H, Martini-Stoica H, Wang B, Yang L, Xu Y, Swartzlander DB, Palmieri M, di Ronza A, Lee VM, Sardiello M, Ballabio A, Zheng H (2014) Selective clearance of aberrant tau proteins and rescue of neurotoxicity by transcription factor EB. *EMBO Mole Med* 6(9):1142–1160. <https://doi.org/10.15252/emmm.201303671>
- Del Prete D, Suski JM, Oulès B, Debayle D, Gay AS, Lacas-Gervais S, Bussiere R, Bauer C, Pinton P, Paterlini-Bréchet P, Wiekowski

- MR, Checler F, Chami M (2016) Localization and processing of the Amyloid- β protein precursor in mitochondria-associated membranes. *J Alzheimer's Dis* 55(4):1549–1570. <https://doi.org/10.3233/JAD-160953>
- Puertollano R, Ferguson SM, Brugarolas J, Ballabio A (2018) The complex relationship between TFEB transcription factor phosphorylation and subcellular localization. *The EMBO Journal* 37(11):98804. <https://doi.org/10.15252/embj.201798804>
- Radeva MY, Kugelmann D, Spindler V, Waschke J (2014) PKA compartmentalization via AKAP220 and AKAP12 contributes to endothelial barrier regulation. *PLoS ONE* 9(9):e106733. <https://doi.org/10.1371/journal.pone.0106733>
- Rai SN, Singh C, Singh A, Singh MP, Singh BK (2020) Mitochondrial dysfunction: a potential therapeutic target to treat Alzheimer's disease. *Mol Neurobiol* 57(7):3075–3088. <https://doi.org/10.1007/s12035-020-01945-y>
- Ramakrishna K, Nalla LV, Naresh D, Venkateswarlu K, Viswanadh MK, Nalluri BN, Chakravarthy G, Duguluri S, Singh P, Rai SN, Kumar A, Singh V, Singh SK (2023) WNT- β Catenin signaling as a potential therapeutic target for neurodegenerative diseases: current status and future perspective. *Diseases (Basel, Switzerland)*, 11(3). <https://doi.org/10.3390/diseases11030089>
- Ramakrishna K, Karuturi P, Siakabinga Q, TAG, Krishnamurthy S, Singh S, Kumari S, Kumar GS, Sobhia ME, Rai SN (2024). Indole-3 Carbinol and Diindolylmethane Mitigated β -Amyloid-Induced Neurotoxicity and Acetylcholinesterase Enzyme Activity: In Silico In Vitro and Network Pharmacology Study. *Diseases (Basel, Switzerland)*. <https://doi.org/10.3390/diseases12080184>
- Ricciarelli R, Brullo C, Prickaerts J, Arancio O, Villa C, Rebosio C, Calcagno E, Balbi M, van Hagen BT, Argyrousi EK, Zhang H (2017) Memory-enhancing effects of GEBR-32a, a new PDE4D inhibitor holding promise for the treatment of Alzheimer's disease. *Sci Rep* 7(1):1–4
- Roesler R, Reolon GK, Maurmann N, Schwartzmann G, Schröder N, Amaral OB, Valvassori S, Quevedo J (2014) A phosphodiesterase 4-controlled switch between memory extinction and strengthening in the hippocampus. *Front Behav Neurosci*. <https://doi.org/10.3389/fnbeh.2014.00091>
- Rose G, Hopper A, De Vivo M, Tehim A (2005) Phosphodiesterase inhibitors for cognitive enhancement. *Curr Pharm des* 11(26):3329–3334. <https://doi.org/10.2174/138161205774370799>
- Del Roso A, Vittorini S, Cavallini G, Donati A, Gori Z, Masini M, Poltera M, Bergamini E (2003) Ageing-related changes in the in vivo function of rat liver macroautophagy and proteolysis. *Exp Gerontol* 38:519–527. [https://doi.org/10.1016/S0531-5565\(03\)00060-8](https://doi.org/10.1016/S0531-5565(03)00060-8)
- Rubinshtein DC, Gestwicki JE, Murphy LO, Klionsky DJ (2007) Potential therapeutic applications of autophagy. *Nat Rev Drug Discov* 6(4):304–312. <https://doi.org/10.1038/nrd2272>
- Saletu B, Anderer P, Fischhof PK, Lorenz H, Barousch R, Böhmer F (1992) EEG mapping and psychopharmacological studies with denbufylline in SDAT and MID. *Biol Psychiat* 32(8):668–681.
- Saunders AM, Strittmatter WJ, Schmechel D, St George-Hyslop PH, Pericak-Vance MA, Joo SH, Rosi BL, Gusella JF, Crapper-MacLachlan DR, Alberts MJ, Hulette C, Crain B, Goldgaber D, Roses AD (1993) Association of apolipoprotein E allele $\epsilon 4$ with late-onset familial and sporadic Alzheimer's disease. *Neurology* 43(8):1467–1467. <https://doi.org/10.1212/WNL.43.8.1467>
- Scheff SW, Price DA, Schmitt FA, Mufson EJ (2006) Hippocampal synaptic loss in early Alzheimer's disease and mild cognitive impairment. *Neurobiol Aging* 27(10):1372–1384. <https://doi.org/10.1016/j.neurobiolaging.2005.09.012>
- Scheltens P, De Strooper B, Kivipelto M, Holstege H, Chételet G, Teunissen CE, Cummings J, van der Flier WM (2021) Alzheimer's disease. *The Lancet* 397(10284):1577–1590. [https://doi.org/10.1016/S0140-6736\(20\)32205-4](https://doi.org/10.1016/S0140-6736(20)32205-4)
- Sebastiani G, Morissette C, Lagacé C, Boulé M, Ouellette M-J, McLaughlin RW, Lacombe D, Gervais F, Tremblay P (2006) The cAMP-specific phosphodiesterase 4B mediates A β -induced microglial activation. *Neurobiol Aging* 27(5):691–701. <https://doi.org/10.1016/j.neurobiolaging.2005.03.024>
- Shanmugam A, Gopal S (2020) Alzheimers awareness among high school students in India. *Am J Geriatr Psychiatry* 28:S70–S72
- Singh M, Agarwal V, Pancham P, Jindal D, Agarwal S, Rai SN, Singh SK, Gupta V (2024) A comprehensive review and androgen deprivation therapy and its impact on Alzheimer's disease risk in older men with prostate cancer. *Degener Neurol Neuromuscul Dis* 14:33–46. <https://doi.org/10.2147/DNND.S445130>
- Son JH, Shim JH, Kim K-H, Ha J-Y, Han JY (2012) Neuronal autophagy and neurodegenerative diseases. *Exp Mol Med* 44(2):89. <https://doi.org/10.3858/emm.2012.44.2.031>
- Spilman P, Podlutska N, Hart MJ, Debnath J, Gorostiza O, Bredeisen D, Richardson A, Strong R, Galvan V (2010) Inhibition of mTOR by rapamycin abolishes cognitive deficits and reduces amyloid- β levels in a mouse model of Alzheimer's disease. *PLoS ONE* 5(4):e9979. <https://doi.org/10.1371/journal.pone.0009979>
- Srivastava P, Tripathi PN, Sharma P, Rai SN, Singh SP, Srivastava RK, Shankar S, Shrivastava SK (2019) Design and development of some phenyl benzoxazole derivatives as a potent acetylcholinesterase inhibitor with anti-oxidant property to enhance learning and memory. *Eur J Med Chem* 163:116–135. <https://doi.org/10.1016/j.ejmech.2018.11.049>
- Steingrímsson E, Copeland NG, Jenkins NA (2004) Melanocytes and the *Microphthalmia* transcription factor network. *Annu Rev Genet* 38(1):365–411. <https://doi.org/10.1146/annurev.genet.38.072902.092717>
- Sugarman MA, McKee AC, Stein TD, Tripodis Y, Besser LM, Martin B, Palmisano JN, Steinberg EG, O'Connor MK, Au R, McClean M, Killiany R, Mez J, Weiner MW, Kowall NW, Stern RA, Alosco ML (2019) Failure to detect an association between self-reported traumatic brain injury and Alzheimer's disease neuropathology and dementia. *Alzheimer's & Dementia* 15(5):686–698. <https://doi.org/10.1016/j.jalz.2018.12.015>
- Sumitomo A, Tomoda T (2021) Autophagy in neuronal physiology and disease. *Curr Opin Pharmacol* 60:133–140. <https://doi.org/10.1016/j.coph.2021.07.013>
- Sun Q, Wei L-L, Zhang M, Li T-X, Yang C, Deng S-P, Zeng Q-C (2019) Rapamycin inhibits activation of AMPK-mTOR signaling pathway-induced Alzheimer's disease lesion in hippocampus of rats with type 2 diabetes mellitus. *Int J Neurosci* 129(2):179–188. <https://doi.org/10.1080/00207454.2018.1491571>
- Switon K, Kotulska K, Janusz-Kaminska A, Zmorzynska J, Jaworski J (2017) Molecular neurobiology of mTOR. *Neuroscience* 341:112–153. <https://doi.org/10.1016/j.neuroscience.2016.11.017>
- Tausif YM, Sai TE, Rajashekarappa RK, Thekkekkara D, Jahagirdar V, Meheronnisha MSK, HariPriya HG, Babu A (2025) Therapeutic potential of plant based HMG CoA reductase inhibitor-berberine for Alzheimer's disease in ICV induced streptozotocin rodent model. *Indian J Pharm Edu Res* 59(1):153–163. <https://doi.org/10.5530/ijper.20255856>
- Thumm M, Egner R, Koch B, Schlumpberger M, Straub M, Veenhuis M, Wolf DH (1994) Isolation of autophagocytosis mutants of *Saccharomyces cerevisiae*. *FEBS Lett* 349(2):275–280. [https://doi.org/10.1016/0014-5793\(94\)00672-5](https://doi.org/10.1016/0014-5793(94)00672-5)
- Tramutola A, Triplett JC, Di Domenico F, Niedowicz DM, Murphy MP, Coccia R, Perluigi M, Butterfield DA (2015) Alteration of mTOR signaling occurs early in the progression of Alzheimer disease (AD): analysis of brain from subjects with pre-clinical

- AD, amnesic mild cognitive impairment and late-stage AD. *J Neurochem* 133(5):739–749. <https://doi.org/10.1111/jnc.13037>
- Tripathi PN, Lodhi A, Rai SN, Nandi NK, Dumoga S, Yadav P, Tiwari AK, Singh SK, El-Shorbagi A-NA, Chaudhary S (2024) Review of pharmacotherapeutic targets in Alzheimer's disease and its management using traditional medicinal plants. *Degener Neurol Neuromus Dis* 14:47–74. <https://doi.org/10.2147/DNND.S452009>
- Tripathi PN, Srivastava P, Sharma P, Tripathi MK, Seth A, Tripathi A, Rai SN, Singh SP, Shrivastava SK (2019) Biphenyl-3-oxo-1,2,4-triazine linked piperazine derivatives as potential cholinesterase inhibitors with anti-oxidant property to improve the learning and memory. *Bioorg Chem* 85:82–96. <https://doi.org/10.1016/j.bioorg.2018.12.017>
- Tsukada M, Ohsumi Y (1993) Isolation and characterization of autophagy-defective mutants of *Saccharomyces cerevisiae*. *FEBS Lett* 333(1–2):169–174. [https://doi.org/10.1016/0014-5793\(93\)80398-e](https://doi.org/10.1016/0014-5793(93)80398-e)
- Um J-H, Park S-J, Kang H, Yang S, Foretz M, McBurney MW, Kim MK, Viollet B, Chung JH (2010) AMP-Activated protein kinase-deficient mice are resistant to the metabolic effects of resveratrol. *Diabetes* 59(3):554–563. <https://doi.org/10.2337/db09-0482>
- Vanmierlo T, Creemers P, Akkerman S, van Duinen M, Sambeth A, De Vry J, Uz T, Blokland A, Prickaerts J (2016) The PDE4 inhibitor roflumilast improves memory in rodents at non-emetic doses. *Behav Brain Res* 303:26–33. <https://doi.org/10.1016/j.bbr.2016.01.031>
- Wang Y, Rijal B, Xu M, Li Z, An Y, Zhang F, Lu C (2020b) Renal denervation improves vascular endothelial dysfunction by inducing autophagy via AMPK/mTOR signaling activation in a rat model of type 2 diabetes mellitus with insulin resistance. *Acta Diabetol* 57(10):1227–1243. <https://doi.org/10.1007/s00592-020-01532-6>
- Wang H, Wang R, Carrera I, Xu S, Lakshmana MK (2016) TFEB overexpression in the P301S model of tauopathy mitigates increased PHF1 levels and lipofuscin puncta and rescues memory deficits. *Eneuro*. <https://doi.org/10.1523/ENEURO.0042-16.2016>
- Wang C, Yang X-M, Zhuo Y-Y, Zhou H, Lin H-B, Cheng Y-F, Xu J-P, Zhang H-T (2012) The phosphodiesterase-4 inhibitor rolipram reverses A β -induced cognitive impairment and neuroinflammatory and apoptotic responses in rats. *Int J Neuropsychopharmacol* 15(06):749–766. <https://doi.org/10.1017/S1461145711000836>
- Wang H, Zhang F, Xu Y, Fu H, Wang X, Wang L, Chen W, Xu X, Gao Y, Zhang J, Zhang H-T (2020a) The phosphodiesterase-4 inhibitor roflumilast, a potential treatment for the comorbidity of memory loss and depression in Alzheimer's Disease: a pre-clinical study in APP/PS1 transgenic mice. *Int J Neuropsychopharmacol* 23(10):700–711. <https://doi.org/10.1093/ijnp/pyaa048>
- Wei W (2002) Abeta 17–42 in Alzheimer's disease activates JNK and caspase-8 leading to neuronal apoptosis. *Brain* 125(9):2036–2043. <https://doi.org/10.1093/brain/awf205>
- Xiao Q, Yan P, Ma X, Liu H, Perez R, Zhu A, Gonzales E, Burchett JM, Schuler DR, Cirrito JR, Diwan A (2014) Enhancing astrocytic lysosome biogenesis facilitates A β clearance and attenuates amyloid plaque pathogenesis. *J Neurosci* 34(29):9607–9620
- Yadav AK, Singh NK, Singh A, Ashish A, Singh S, Rai SN, Singh SK, Singh R (2024) Investigation of serum pro-inflammatory markers and trace elements among short stature in eastern Uttar Pradesh and Bihar populations. *J Inflamm Res* 17:6063–6073. <https://doi.org/10.2147/JIR.S473895>
- Yang C, Su C, Iyaswamy A, Krishnamoorthi SK, Zhu Z, Yang S, Tong BC, Liu J, Sreenivasmurthy SG, Guan X, Kan Y, Wu AJ, Huang AS, Tan J, Cheung K, Song J, Li M (2022) Celastrol enhances transcription factor EB (TFEB)-mediated autophagy and mitigates Tau pathology: Implications for Alzheimer's disease therapy. *Acta Pharmaceutica Sinica B* 12(4):1707–1722. <https://doi.org/10.1016/j.apsb.2022.01.017>
- Yorimitsu T, Klionsky DJ (2005) Autophagy: molecular machinery for self-eating. *Cell Death Differ* 12(S2):1542–1552. <https://doi.org/10.1038/sj.cdd.4401765>
- Zhang L, Fang Y, Zhao X, Zheng Y, Ma Y, Li S, Huang Z, Li L (2021) BRUCE silencing leads to axonal dystrophy by repressing autophagosome-lysosome fusion in Alzheimer's disease. *Transl Psychiatry* 11(1):421. <https://doi.org/10.1038/s41398-021-01427-2>
- Zhang S, Salemi J, Hou H, Zhu Y, Mori T, Giunta B, Obregon D, Tan J (2010) Rapamycin promotes β -amyloid production via ADAM-10 inhibition. *Biochem Biophys Res Commun* 398(3):337–341. <https://doi.org/10.1016/j.bbrc.2010.06.017>
- Zhang J, Wang J, Zhou Z, Park J-E, Wang L, Wu S, Sun X, Lu L, Wang T, Lin Q, Sze SK, Huang D, Shen H-M (2018) Importance of TFEB acetylation in control of its transcriptional activity and lysosomal function in response to histone deacetylase inhibitors. *Autophagy* 14(6):1–17. <https://doi.org/10.1080/15548627.2018.1447290>
- Zheng L, Terman A, Hallbeck M, Dehvari N, Cowburn RF, Benedikz E, Kågedal K, Cedazo-Minguez A, Marcusson J (2011) Macroautophagy-generated increase of lysosomal amyloid β -protein mediates oxidant-induced apoptosis of cultured neuroblastoma cells. *Autophagy* 7(12):1528–1545. <https://doi.org/10.4161/auto.7.12.18051>

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